EXP.NO:16

DP - PLAYING WITH NUMBERS

PROBLEM STATEMENT:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

CODE:

```c
#include<stdio.h>

int main(){
    int n;
    scanf("%d", &n);
    long long dp[n+1];
    dp[0] = 1;
    for(int i=1;i<=n;i++){
        dp[i]=dp[i-1];
        if(i>=3){
            dp[i]+=dp[i-3];
        }
    }
    printf("%lld", dp[n]);
    return 0;
}
```

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OUTPUT:



	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

RESULT:

The code is executed successfully.